A Three-Stage Quantitative Research Pipeline: Signal Discovery, Strategy Construction, and Portfolio Execution

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Project Overview

This project overview summarizes a three-stage quantitative finance project built on daily ETF data from 2014 through 2025. The project begins with signal and model discovery, where volatility forecasting, cointegration, Ornstein–Uhlenbeck (OU) spread modeling, and Hidden Markov Model (HMM) regime detection are tested as research hypotheses. It then converts selected signals into trading rules: volatility timing on SPY, pairs trading on EFA–EEM, and regime switching on SPY. Finally, the retained strategy returns are combined into portfolio-level allocations using inverse-volatility weighting, Markowitz mean–variance optimization, volatility targeting, transaction costs, slippage, turnover caps, and robustness diagnostics. The main result is about research discipline rather than claiming a ready-to-trade system: volatility and regime signals produce useful risk-management behavior, while the EFA–EEM pair fails the cointegration test and performs poorly when traded. At the portfolio level, inverse-volatility weighting is the most robust selected allocator after execution costs, while Markowitz variants require regularization penalties or hard caps to reduce noisy reallocations. The project therefore demonstrates a central lesson of quantitative research: statistical structure, tradable strategy performance, and realistic portfolio usefulness are related but distinct questions.

1 Introduction

This project develops a complete research pipeline for testing systematic trading ideas. The purpose is not to claim a production-ready trading system, but to show how a quantitative idea should be examined layer by layer. A model can look statistically meaningful and still fail after

transaction costs. A strategy can perform well alone and still add little value to a portfolio. A portfolio optimizer can look strong before costs and fragile after realistic execution assumptions.

The work is organized into three stages:

1. **Signal and model discovery**: test whether market structure exists in volatility, cross-asset spreads, and regimes.
2. **Strategy construction**: convert research signals into positions and backtested returns.
3. **Portfolio and execution simulation**: combine the strategies, apply risk targets, include costs, and evaluate robustness.

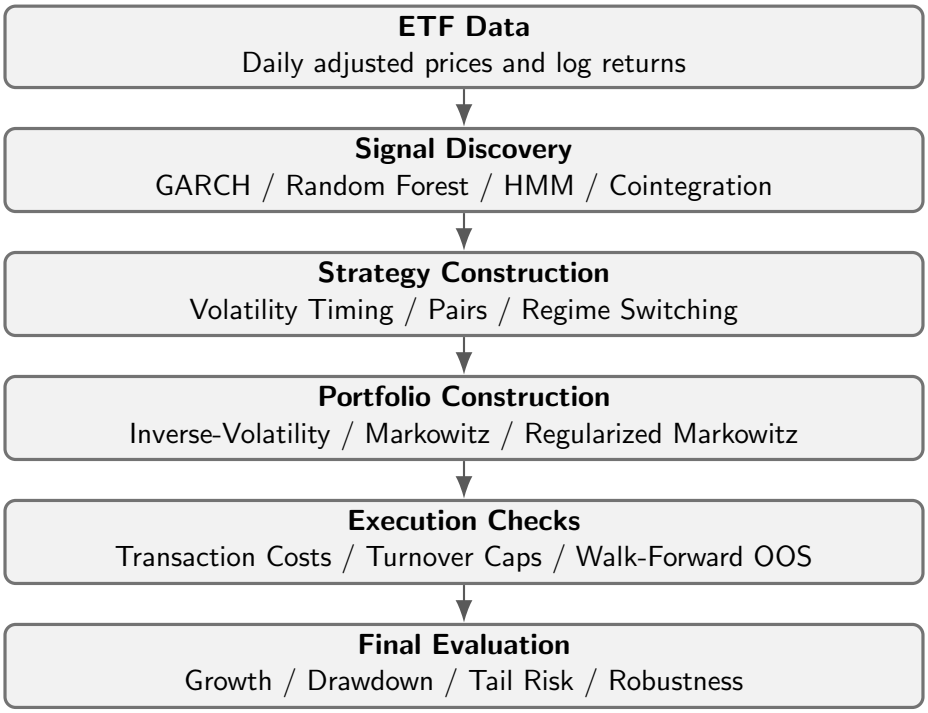


Figure 1: End-to-end project workflow. Each stage narrows the set of ideas before the final portfolio evaluation.

The analysis uses adjusted close prices for eight ETFs: SPY, TLT, GLD, USO, UUP, EFA, EEM, and FXE. These assets provide a simple multi-asset universe covering U.S. equities, long-duration Treasuries, gold, oil, the U.S. dollar, developed international equities, emerging markets, and the euro. The configured sample begins on 2014-01-01 and ends on 2025-12-31.

Unless otherwise stated, all rolling calculations use daily observations, the annualization factor is 252, the main benchmark is SPY because it tracks the S&P 500, a broad index of large U.S. companies, making it a liquid proxy for general U.S. equity-market risk, the configured pair is EFA–EEM, the transaction-cost assumption is 5 basis points, the slippage assumption is 2 basis points, the portfolio turnover cap is 0.35, and the final target portfolio volatility is 10% annualized.

2 Risk Metrics and Interpretation

The project evaluates every strategy and portfolio with the same performance and risk vocabulary. Let $\{r_t\}_{t=1}^{T}$ denote daily strategy or portfolio returns, let T be the number of observations, and let $A = 252$ be the annualization factor. The sample mean and sample standard deviation are

$$\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t, \qquad s_r = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2}. \tag{1}$$

Sharpe Ratio

The annualized Sharpe ratio is

$$\text{Sharpe} = \sqrt{A}\frac{\bar{r}}{s_r}, \tag{2}$$

when $s_r > 0$. Sharpe measures average excess reward per unit of total volatility. In this project the risk-free rate is omitted because the focus is on comparing internally generated return streams under the same daily-return convention, so the numerator is the average strategy return itself.

Sortino Ratio

The Sortino ratio is similar, but it penalizes downside volatility rather than all volatility. Let $\mathcal{T}^- = \{t : r_t < 0\}$ and let s_r^- be the standard deviation of the negative-return subsample $\{r_t : r_t < 0\}$. Then

$$\text{Sortino} = \sqrt{A}\frac{\bar{r}}{s_r^-}, \tag{3}$$

when $s_r^- > 0$. This is useful because upside and downside variation are not equally undesirable: a strategy that has large positive jumps should not be penalized in the same way as a strategy with large losses.

Compound Growth and CAGR

Compound growth is tracked through the equity curve,

$$E_t = \prod_{u=1}^{t}(1 + r_u), \tag{4}$$

where $E_0 = 1$ and E_t is the growth of one dollar after linking daily returns. If $Y = T/A$ is the sample length in years, the compound annual growth rate is

$$\text{CAGR} = E_T^{1/Y} - 1. \tag{5}$$

CAGR is reported because it measures realized compounded growth rather than one-period average return.

Maximum Drawdown and Calmar Ratio

Maximum drawdown measures the worst peak-to-trough loss along the realized path. With the running peak $\max_{u \le t} E_u$, maximum drawdown is

$$\mathrm{MDD} = \min_{t \le T} \left(\frac{E_t}{\max_{u \le t} E_u} - 1 \right). \tag{6}$$

It is negative by construction in a losing episode. Drawdown is path-dependent: two strategies can have the same final CAGR and volatility but very different investor experience if one spends more time below its previous high. The Calmar ratio compares compounded growth to the magnitude of this worst loss,

$$\mathrm{Calmar} = \frac{\mathrm{CAGR}}{|\mathrm{MDD}|}, \tag{7}$$

when $\mathrm{MDD} < 0$. A higher Calmar ratio means that the strategy produced more annual growth per unit of worst historical drawdown.

Annualized Volatility

Annualized volatility is reported as

$$\sigma_{\mathrm{ann}} = s_r \sqrt{A}, \tag{8}$$

which is the same scale used in the Sharpe denominator and in the portfolio volatility-targeting step. Volatility is not a complete risk measure because it treats upside and downside deviations symmetrically, but it is still useful for sizing positions, comparing strategies, and targeting the final portfolio to 10% annualized volatility.

Value at Risk

Tail risk is measured with historical Value at Risk and Expected Shortfall. Let R denote a generic daily return, let $F_R(u) = \mathbb{P}(R \le u)$ be its cumulative distribution function, and let $c \in (0, 1)$ be a confidence level. Define $\alpha = 1 - c$, so $c = 0.95$ implies $\alpha = 0.05$ and focuses attention on the worst 5% of daily returns. The theoretical left-tail quantile is

$$q_\alpha = F_R^{-1}(\alpha) = \inf\{u : F_R(u) \ge \alpha\}. \tag{9}$$

Because bad daily returns are negative, the project reports VaR as a positive loss magnitude:

$$\mathrm{VaR}_c = -q_\alpha. \tag{10}$$

In the historical implementation, q_α is replaced by the empirical quantile $\hat{q}_\alpha$, so

$$\widehat{\mathrm{VaR}}_c = -\hat{q}_\alpha. \tag{11}$$

A reported daily VaR_{95} of 1% therefore means that the empirical worst 5% threshold is approximately a 1% loss.

Expected Shortfall

Expected Shortfall, also called Conditional VaR, measures the average severity inside the same left tail rather than only the tail threshold. The theoretical definition is

$$\mathrm{ES}_c = -\mathbb{E}\left[R \mid R \le q_\alpha\right], \tag{12}$$

and the empirical version used here is

$$\widehat{\mathrm{ES}}_c = -\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} r_t, \qquad \mathcal{T} = \{t : r_t \le \hat{q}_\alpha\}. \tag{13}$$

Expected Shortfall is usually at least as large as VaR under the same tail convention because it averages the losses beyond the VaR cutoff. This makes ES especially helpful when two strategies have similar VaR but one has much worse losses in the deepest part of the tail.

Reading the Metrics Together

These metrics answer different questions and should be read together. Sharpe and Sortino summarize average risk-adjusted performance; CAGR summarizes compounded growth; maximum drawdown and Calmar describe the realized path and recovery burden; volatility describes scale; and VaR and Expected Shortfall describe bad-day severity. The later strategy and portfolio tables use this common language so that model quality, tradable strategy performance, and realistic portfolio usefulness are evaluated on the same footing.

3 Data and Research Design

Let S_t denote an adjusted asset price at time t. The project uses daily log returns,

$$r_t = \log\left(\frac{S_t}{S_{t-1}}\right), \tag{14}$$

because log returns are time-additive and convenient for volatility and regime modeling. Rolling realized volatility over a window of m trading days is estimated as

$$\hat{\sigma}_t^{\text{realized}} = \sqrt{252}\,\mathrm{Std}(r_{t-m+1}, \ldots, r_t), \tag{15}$$

where 252 is the annualization factor used for daily trading data. In the implementation, $m = 21$, so realized volatility is based on roughly one trading month of returns.

The project keeps one shared data foundation across all notebooks. This is important because otherwise differences between model results, strategy results, and portfolio results could be caused by inconsistent calendars rather than genuine modeling effects. The codebase separates the work

into reusable modules for data loading, volatility modeling, OU estimation, regime modeling, strategy construction, backtesting, portfolio weighting, and execution-cost accounting.

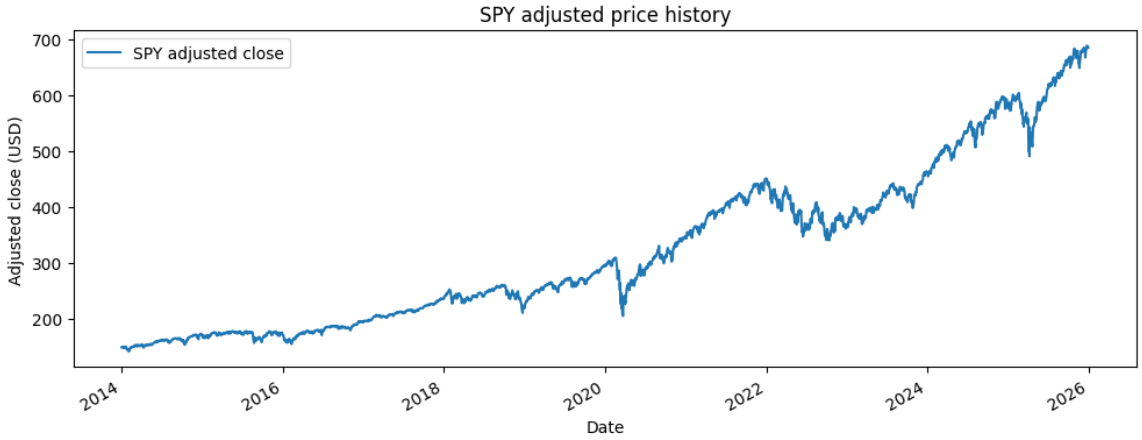


Figure 2: SPY adjusted price history used as the main benchmark for volatility timing and regime switching. The long upward trend is interrupted by drawdowns, which motivates risk-aware overlays rather than constant exposure.

4 Signal and Model Discovery

4.1 Volatility Forecasting

The first research question is whether next-period risk is predictable. Two models are compared. The first is a GARCH$(1,1)$ model,

$$r_t = \sigma_t \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1), \tag{16}$$

$$\underbrace{\sigma_t^2}_{\text{today variance forecast}} = \omega + \underbrace{\alpha r_{t-1}^2}_{\text{reaction to yesterday's shock}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{volatility persistence}}. \tag{17}$$

Here r_t is the SPY log return on day t, σ_t is the conditional volatility known from past information, and ω is the baseline variance level. This captures volatility clustering: large shocks tend to be followed by higher future volatility. In the code, this model is fit as a zero-mean GARCH$(1,1)$ on SPY daily returns after multiplying returns by 100 for numerical stability; the fitted conditional volatility is then divided by 100 to return to decimal units.

The second model is a random forest forecast,

$$\hat{\sigma}_{t+1} = f(x_t), \tag{18}$$

where x_t contains lagged returns, rolling moments, and current realized volatility features. The feature set uses return lags of 1 and 5 trading days, a 63-day rolling mean, standard deviation, skewness, and kurtosis, and the current 21-day realized volatility. The target is the next available

realized-volatility value, so the forecast is one step ahead. It uses a chronological $70\%/30\%$ train/test split, and the random forest uses 300 trees and maximum depth 4, to prevent overfitting.

Model	MSE	MAE	Correlation
Random Forest	0.000188	0.008328	0.985481
GARCH$(1,1)$	0.027052	0.136491	0.901712

Table 1: Volatility forecast comparison. Lower MSE and MAE indicate smaller forecast errors.

The table shows higher correlation which means closer co-movement or that the forecasts move closely with realized volatility over time. This is why the results point to persistent volatility: models that use yesterday's variance, recent realized volatility, and lagged return features can predict future realized volatility well. Moreover, both models track realized volatility, but the random forest fits more tightly because it uses recent realized-volatility information and additional lagged features. This is a risk forecast, not a directional return forecast: it says more about how risky tomorrow may be than whether SPY will rise or fall.

4.2 Cointegration and OU Spread Modeling

The second research question is whether a relative-value spread can be traded. For two assets $P_t^{(1)}$ and $P_t^{(2)}$, the spread is defined as

$$X_t = P_t^{(1)} - \alpha - \beta P_t^{(2)}, \tag{19}$$

where β is the hedge ratio estimated by ordinary least squares. In the configured pair study, $P_t^{(1)}$ is EFA and $P_t^{(2)}$ is EEM, both measured with adjusted close prices on the shared calendar. In regression form,

$$P_t^{(1)} = \alpha + \beta P_t^{(2)} + u_t, \tag{20}$$

so the residual u_t is the estimated spread. The Engle–Granger procedure tests whether this residual is stationary. If it is not stationary, the pair should not be treated as reliably mean-reverting.

The spread dynamics are interpreted through an OU process,

$$dX_t = \kappa(\theta - X_t)dt + \sigma_X dW_t, \tag{21}$$

where κ is the speed of mean reversion, θ is the long-run mean, and σ_X is the spread volatility. The project reports both the default ΔX_t regression estimate and an exact discrete-time OU/AR(1) estimate with daily step size $\Delta t = 1$; the quoted half-life for EFA–EEM uses the exact discrete estimate. The half-life of mean reversion is

$$\tau_{1/2} = \frac{\ln 2}{\kappa}. \tag{22}$$

The standardized trading signal is the rolling z-score,

$$z_t = \frac{X_t - \bar{X}_t}{s_t}, \tag{23}$$

where $\bar{X}_t$ and s_t are rolling estimates of the spread mean and standard deviation. In this project the z-score window is 21 trading days, the same one-month window used for realized volatility. This is the setting used for the spread plot and for the EFA–EEM trading rule.

The broad ETF-pair scan found that 0 of 28 pairs rejected the no-cointegration null at the 5% level. The configured EFA–EEM pair also failed the cointegration test, with test statistic -1.734324, p-value 0.661040, and 5% critical value -3.338157. Although the exact discrete OU fit produced a small positive mean-reversion estimate $\kappa \approx 0.002496$, corresponding to a half-life of about 278 trading days, the cointegration result warns that this is not a robust statistical-arbitrage candidate.

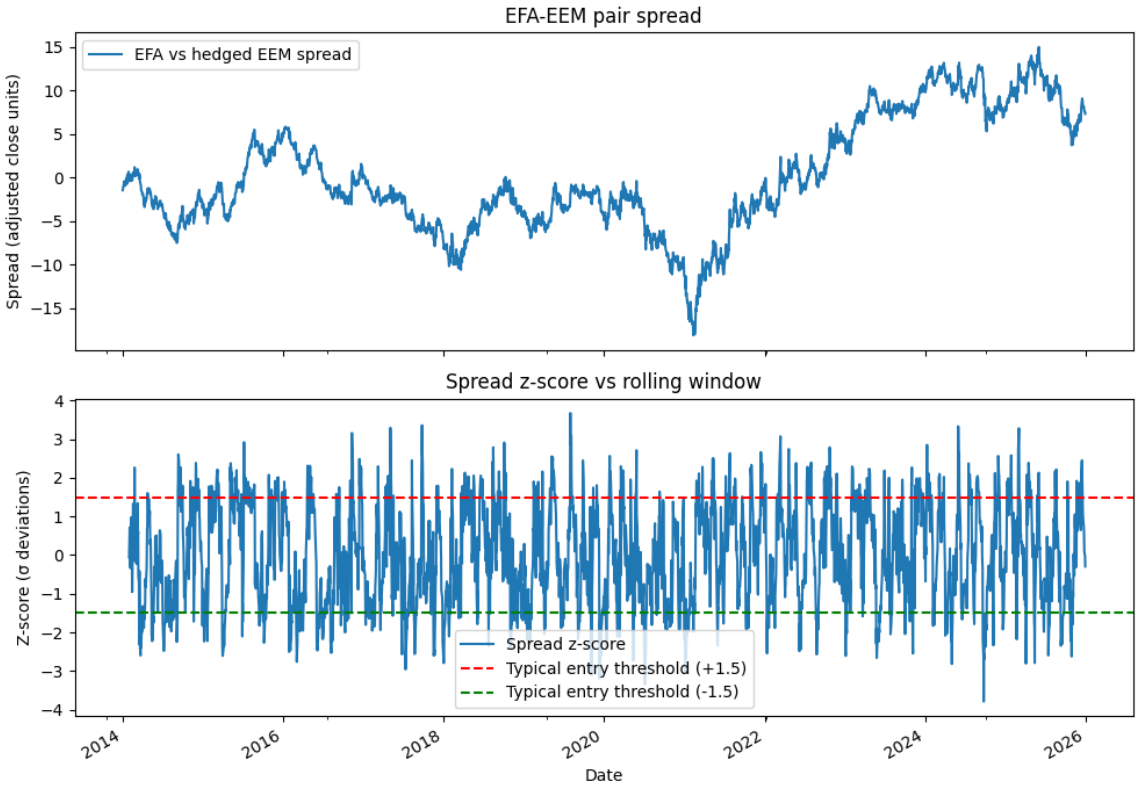


Figure 3: EFA–EEM spread and z-score. The strategy enters when the spread is unusually high or low, but the failed cointegration evidence suggests that these deviations are not reliable mean-reversion opportunities.

4.3 Hidden Markov Model Regime Detection

The third research question is whether market regimes can be inferred from returns, realized volatility, and momentum. The implemented HMM uses $K = 3$ regimes so the states can be

interpreted as bull, sideways, and crisis-like environments. Let $Z_t \in \{1, \ldots, K\}$ be an unobserved regime. The HMM assumes

$$\mathbb{P}(Z_t = j \mid Z_{t-1} = i) = A_{ij}, \tag{24}$$

where A is the transition matrix. Conditional on the regime, the observed feature vector is assumed to be Gaussian:

$$Y_t \mid Z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \tag{25}$$

with

$$Y_t = \begin{bmatrix} r_t \\ \hat{\sigma}_t^{\text{realized}} \\ \text{momentum}_t \end{bmatrix}. \tag{26}$$

Here r_t is the daily SPY log return, $\hat{\sigma}_t^{\text{realized}}$ is the 21-day annualized realized volatility, and momentum_t is the 21-day rolling mean of SPY returns. The feature vector is assumed to be Gaussian within each regime so the HMM has a simple probability model for judging how likely the observed return, volatility, and momentum pattern is under each hidden state. Before fitting, these features are standardized so variables with larger numerical scales do not dominate the regime classification. The Gaussian HMM uses full covariance matrices, meaning it allows return, volatility, and momentum to move together within each regime with up to 500 expectation–maximization iterations.

Regime	Return Mean	Return Std.	Realized Vol. Mean	Count
0	0.000866	0.005373	0.086501	1408
1	0.000278	0.010578	0.161927	1249
2	-0.000033	0.023033	0.334244	339

Table 2: HMM regime summary. Return Mean and Return Std. are daily log-return statistics reported in decimal form, while Realized Vol. Mean is the mean 21-day annualized realized volatility; multiplying by 100 converts these values to percentages. Regime 0 is calm and positive, regime 1 is intermediate, and regime 2 is crisis-like.

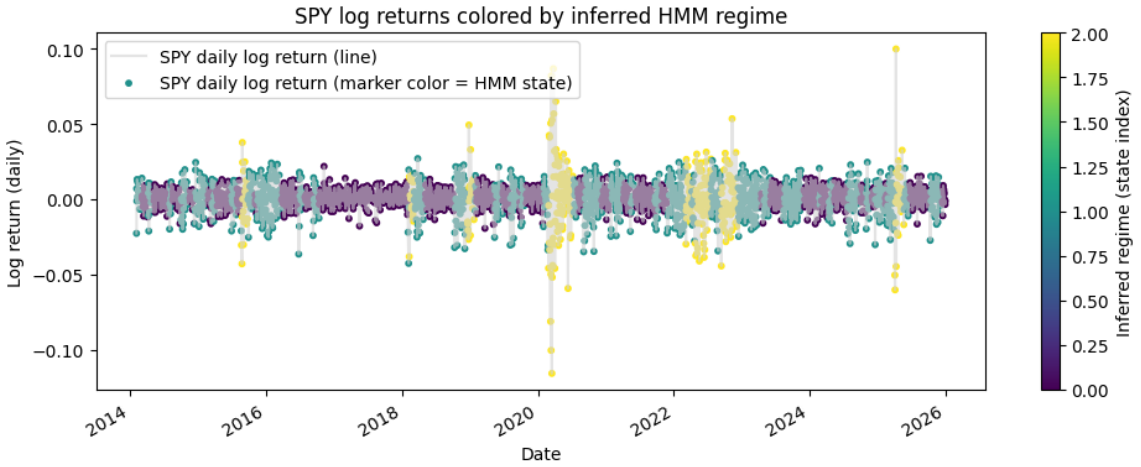


Figure 4: Daily SPY log returns plotted against date and colored by the inferred HMM regime. The figure shows how the HMM separates calm low-volatility periods from high-volatility stress periods in the return series.

The HMM produces economically interpretable states, and the regime counts show where the market spent most of the research sample. Regime 0 is the most common state, appearing on $1{,}408$ days; it has the highest average return and lowest realized volatility, so it behaves like a calm/bull regime. Regime 1 appears on $1{,}249$ days, making it nearly as common, but it has higher volatility and only moderate positive returns, so it represents a normal intermediate or sideways regime. Regime 2 appears on only 339 days, so crisis-like conditions are relatively rare; however, this state is the most important for risk management because it has near-zero or negative average return and much higher volatility. Overall, the research sample is mostly calm or moderate, with shorter but more severe stress episodes.

These discovery results now become the inputs for the trading layer. The volatility forecast becomes the signal for scaling SPY exposure in the volatility-timing strategy, the EFA–EEM spread z-score becomes the entry and exit signal for the pairs-trading strategy, and the HMM regime label becomes the signal for reducing or increasing SPY exposure in the regime-switching strategy. Stating the transition this way is important because the next section does not introduce unrelated trades; it tests whether the three research signals identified above can survive the move from statistical evidence to actual portfolio positions.

5 Strategy Construction

The second stage converts research signals into positions and applies one consistent backtesting engine. If w_t is the portfolio weight vector chosen at time t, r_{t+1} is the next-period asset-return vector, and the subscript p denotes the resulting portfolio, the gross return is

$$r_{p,t+1}^{\text{gross}} = w_t^{\top} r_{t+1}. \tag{27}$$

Turnover is

$$\tau_t = \sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}|, \tag{28}$$

and the net return after transaction costs and slippage is

$$r_{p,t+1}^{\text{net}} = r_{p,t+1}^{\text{gross}} - \tau_t \frac{c}{10{,}000}, \tag{29}$$

where c is the cost in basis points per unit of turnover. In all strategy backtests here, positions are lagged by one trading day before earning returns, and $c = 7$ basis points because the project combines 5 basis points of transaction cost with 2 basis points of slippage. Thus a one-unit turnover event is charged $7/10{,}000 = 0.0007$ in return units.

5.1 Volatility Timing

The volatility-timing strategy uses the predicted volatility to scale SPY exposure:

$$w_t^{\text{VT}} \propto \frac{1}{\hat{\sigma}_{t+1}}. \tag{30}$$

The intuition is simple: when predicted risk is high, hold less SPY; when predicted risk is low, hold more SPY. In the implementation, the raw inverse-volatility weight is normalized by its expanding average so the exposure stays on a stable scale, then clipped between 0 and a maximum leverage of 1.5, meaning the strategy cannot short SPY and cannot take more than 150% SPY exposure. This strategy does not attempt to forecast returns directly. It attempts to improve the path of returns by controlling exposure to forecasted risk.

To make the test meaningful, the volatility-timing rule is compared with a naive buy-and-hold (BH) SPY benchmark. Writing SPY as a one-asset return vector, the benchmark keeps constant exposure,

$$\boldsymbol{w}_t^{\text{BH}} = \begin{bmatrix} 1 \end{bmatrix}. \tag{31}$$

The volatility-timing (VT) strategy earns the following net return after trading costs,

$$r_{t+1}^{\text{VT}} = (\boldsymbol{w}_t^{\text{VT}})^\top \boldsymbol{r}_{t+1}^{\text{SPY}} - \tau_t^{\text{VT}} \frac{c}{10{,}000}. \tag{32}$$

The equivalent buy-and-hold net return is

$$\begin{aligned} r_{t+1}^{\text{BH}} &= (\boldsymbol{w}_t^{\text{BH}})^\top \boldsymbol{r}_{t+1}^{\text{SPY}} - \tau_t^{\text{BH}} \frac{c}{10{,}000} \\ &= r_{t+1}^{\text{SPY}}, \qquad \tau_t^{\text{BH}} = 0. \end{aligned} \tag{33}$$

This comparison asks whether dynamic risk scaling improves the return path relative to simply holding SPY every day. The goal is not only higher final wealth, but also better risk-adjusted performance: a higher Sharpe ratio, lower drawdown, and smoother tail-risk profile after trading costs.

5.2 Pairs Trading

The pairs strategy uses the spread z-score rule

$$\text{signal}_t = \begin{cases} -1, & z_t > z_{\text{entry}}, \\ +1, & z_t < -z_{\text{entry}}, \\ 0, & |z_t| < z_{\text{exit}}. \end{cases} \tag{34}$$

If the spread is too high, the strategy shorts the spread; if the spread is too low, it buys the spread. The implemented thresholds are $z_{\text{entry}} = 1.5$ and $z_{\text{exit}} = 0.5$, so the rule has five zones: enter or stay long the spread when $z_t < -1.5$; hold an existing long-spread position when $-1.5 \leq z_t < -0.5$; exit to cash when $-0.5 \leq z_t \leq 0.5$; hold an existing short-spread position when $0.5 < z_t \leq 1.5$; and enter or stay short the spread when $z_t > 1.5$. Positions are equal and opposite across EFA and EEM: a long-spread signal holds $+1$ in EFA and -1 in EEM, while a short-spread signal holds -1 in EFA and $+1$ in EEM. The economic idea is convergence. However, because the EFA–EEM pair failed the cointegration test, this strategy is expected to struggle.

5.3 Regime Switching

The regime-switching strategy maps HMM states into SPY exposure:

$$w_t^{\text{regime}} = \begin{cases} 1.00, & L_t = \text{bull}, \\ 0.35, & L_t = \text{sideways}, \\ 0.00, & L_t = \text{crisis}. \end{cases} \tag{35}$$

After the HMM assigns each day to a latent state, the states are ranked by average SPY return and labeled bull, sideways, or crisis from highest to lowest. The exposures above are then applied as fixed, illustrative risk budgets: full exposure in bull markets, partial exposure in sideways markets, and no exposure in crisis-like markets. Thus, the HMM estimates the regime labels, while the allocation weights are preset policy choices rather than model-estimated or backtest-optimized parameters.

Metric	Vol. Timing	Buy-and-Hold	Pairs	Regime
Sharpe ratio	1.18	1.15	-0.34	1.21
Sortino ratio	1.68	1.60	-0.36	1.41
CAGR (%)	20.63	19.63	-2.95	9.33
Max drawdown (%)	-17.23	-19.21	-16.53	-7.44
Calmar ratio	1.20	1.02	-0.18	1.25
Volatility (%)	17.10	16.75	7.99	7.60
VaR_{95} (%)	1.72	1.57	0.79	0.67
ES_{95} (%)	2.42	2.31	1.32	1.17

Table 3: Single-strategy performance. Values are net of the configured transaction cost and slippage assumptions.

As shown in Table 3 and Fig. 5, after applying transaction costs and slippage, the net strategy results match the research evidence. Volatility timing modestly improves Sharpe, Sortino, CAGR, max drawdown, and Calmar ratio relative to naive buy-and-hold, although its volatility and tail-loss measures are slightly higher. Pairs trading loses money and suffers a roughly -17% max drawdown, consistent with the failed cointegration test.

Regime switching gives the most defensive return path: it has the highest Sharpe ratio (1.21) and Calmar ratio (1.25), the lowest volatility (7.60%), the smallest drawdown (-7.44%), and the lowest VaR_{95} (0.67%) and ES_{95} (1.17%) among the reported strategies. The tradeoff is lower annualized growth: its CAGR is 9.33%, compared with 20.63% for volatility timing and 19.63% for buy-and-hold, because the regime rule deliberately cuts SPY exposure during defensive states. Overall, the table suggests that volatility timing is the better growth overlay, regime switching is the better risk-control overlay, and the EFA–EEM pairs strategy should not be trusted without stronger mean-reversion evidence.

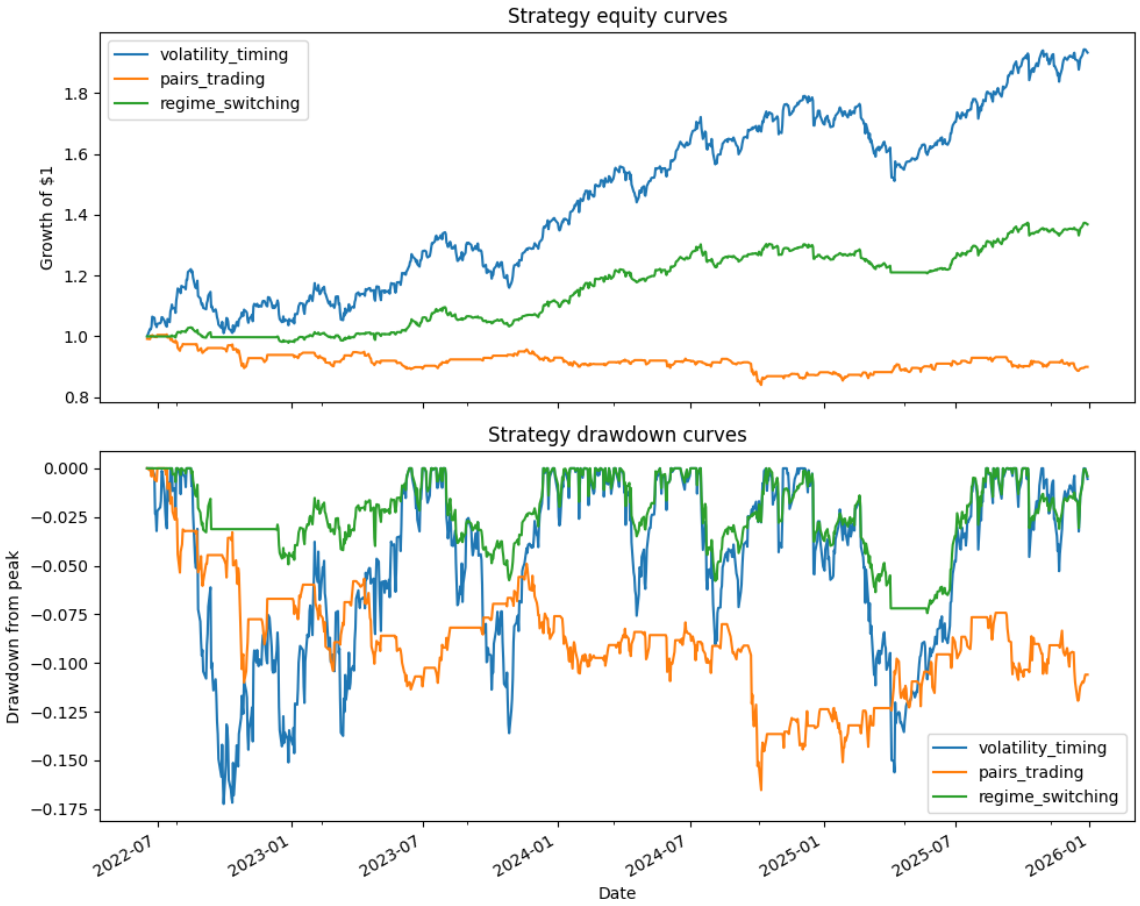


Figure 5: Equity and drawdown curves for the three strategy prototypes. Volatility timing and regime switching produce smoother growth than the weak pairs-trading leg.

5.4 Walk-Forward Out-of-Sample Check

The walk-forward check tests the signals under a real-time discipline. Models are estimated on an expanding history that begins with 590 daily observations, then applied to the next 21 trading days; after each block, the history expands and the process repeats on the same calendar for all three strategies. This is stricter than a single full-sample backtest because each position is generated only from information available before that test block.

Metric	Vol. Timing OOS	Buy-and-Hold OOS	Pairs OOS	Regime OOS
Sharpe ratio	0.92	0.76	-0.24	0.83
Sortino ratio	1.13	0.89	-0.27	0.90
CAGR (%)	10.02	13.06	-2.50	5.99
Max drawdown (%)	-15.26	-35.75	-28.99	-11.60
Calmar ratio	0.66	0.37	-0.09	0.52
VaR_{95} (%)	1.08	1.71	0.95	0.68
ES_{95} (%)	1.76	2.85	1.43	1.18

Table 4: Walk-forward out-of-sample strategy performance using expanding training windows and 21-day test blocks, with naive buy-and-hold included as the benchmark.

As shown in Table 4 and Fig. 6, the stricter test reduces strategy performance and clarifies the benchmark tradeoff. Naive buy-and-hold has the highest OOS CAGR at 13.06%, but it also has the largest max drawdown (-35.75%) and the highest ES_{95} (2.85%). Volatility timing gives up some growth, with CAGR 10.02%, but improves risk-adjusted performance with Sharpe 0.92, Sortino 1.13, and a much smaller max drawdown of -15.26% by scaling exposure down when forecast risk rises.

Regime switching has a lower CAGR of 5.99% because the rule intentionally moves to partial or zero exposure in defensive regimes, which sacrifices some rebound and bull-market upside. The same exposure cuts explain its smaller drawdown (-11.60%) and lower tail risk, with VaR_{95} of 0.68% and ES_{95} of 1.18%. The pairs strategy fails the robustness check, with Sharpe -0.24, CAGR -2.50%, and max drawdown -28.99%, reinforcing the earlier warning from the failed cointegration test. In the figure, this appears as a faster but bumpier volatility-timing curve, a slower but smoother regime-switching curve, and a deteriorating pairs-trading curve.

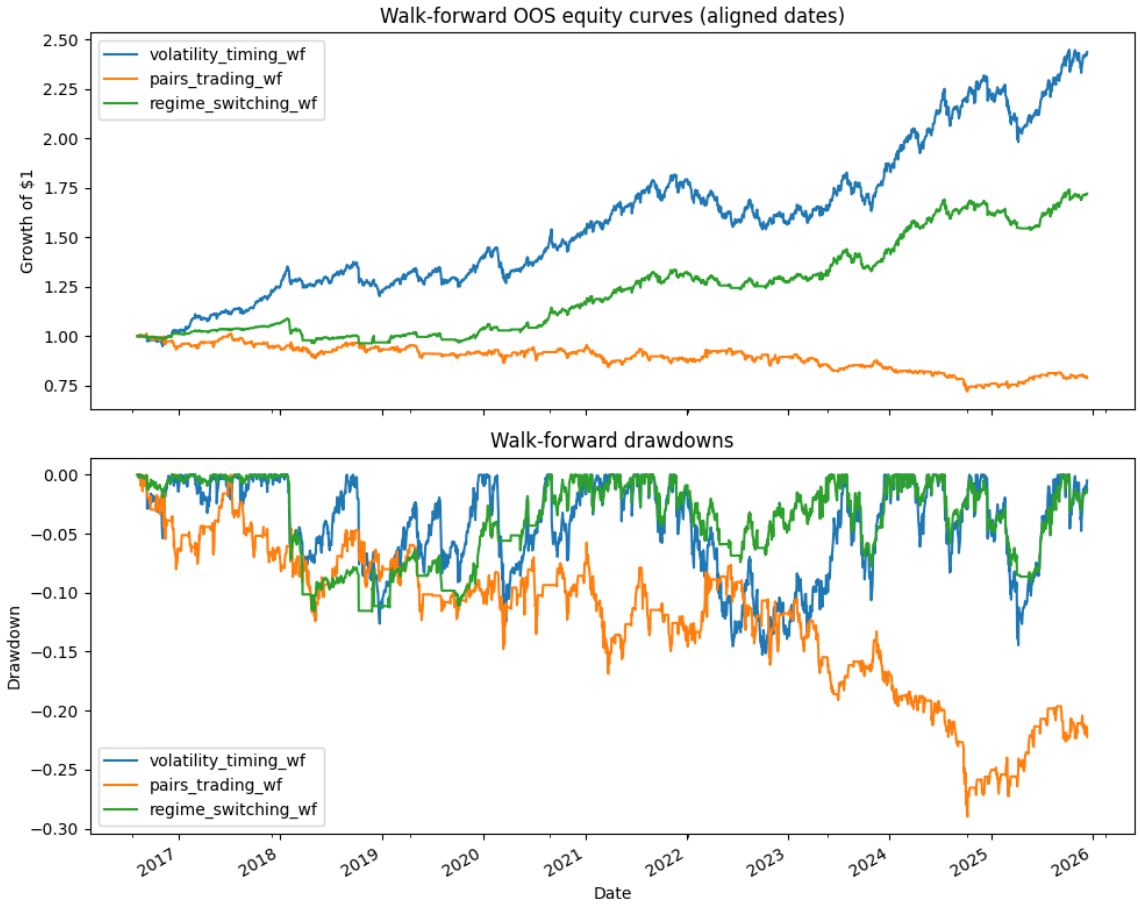


Figure 6: Walk-forward out-of-sample equity and drawdown curves. Volatility timing maintains the best growth profile, regime switching remains more defensive, and pairs trading continues to deteriorate.

6 Portfolio and Execution Simulation

The third stage treats the retained strategy returns as portfolio building blocks. Because the EFA–EEM pairs strategy fails the research and backtest filters, the portfolio layer focuses on the two SPY-based overlays that remain useful: volatility timing and regime switching. In this section, $M = 2$ and the strategies produce returns $r_t^{(1)}, \ldots, r_t^{(M)}$. The portfolio return is

$$r_t^{\text{portfolio}} = \sum_{m=1}^{M} w_{m,t} r_t^{(m)}. \tag{36}$$

6.1 Inverse-Volatility Weighting

Inverse-volatility weighting assigns larger weights to lower-volatility strategies:

$$\tilde{w}_{m,t} = \frac{1}{\hat{\sigma}_{m,t}}, \qquad w_{m,t} = \frac{\tilde{w}_{m,t}}{\sum_{j=1}^{M} \tilde{w}_{j,t}}. \tag{37}$$

Here $\hat{\sigma}_{m,t}$ is the trailing 63-day standard deviation of strategy m's daily returns, clipped below at 10^{-6} to avoid division by zero. This rule is deliberately simple: it uses realized risk, not expected return. In this application, that simplicity is helpful because the lower-volatility regime-switching sleeve receives a larger average allocation, while the higher-growth volatility-timing sleeve still contributes upside.

6.2 Markowitz Optimization

The Markowitz baseline estimates expected returns and covariance from rolling strategy returns and solves a long-only maximum-Sharpe allocation:

$$\max_{w_t} \frac{w_t^\top \hat{\mu}_t}{\sqrt{w_t^\top \hat{\Sigma}_t w_t}}, \tag{38}$$

subject to

$$\sum_{m=1}^{M} w_{m,t} = 1, \qquad 0 \le w_{m,t} \le 1. \tag{39}$$

The implementation estimates $\hat{\mu}_t$ and $\hat{\Sigma}_t$ from a 252-trading-day rolling window, annualizes both with 252, and then uses a ridge-adjusted covariance matrix,

$$\hat{\Sigma}_t^{\text{ridge}} = \hat{\Sigma}_t + \lambda_{\text{ridge}} I, \qquad \lambda_{\text{ridge}} = 10^{-6}, \tag{40}$$

where I is the identity matrix. This small adjustment slightly increases the diagonal entries, increases the covariance matrix eigenvalues, and helps prevent near-singularity when rolling strategy returns are noisy or highly collinear. The optimizer then constrains weights to be long-only and fully invested and solves the allocation problem with SLSQP.

The robustness comparison also tests three Markowitz variants around this baseline: Ledoit–Wolf covariance shrinkage, a hard 70% maximum-weight cap, and a smooth concentration penalty (regularized). The selected regularized Markowitz version keeps the same long-only, fully invested constraints but penalizes concentration:

$$\max_{w_t} \left\{ \frac{w_t^\top \hat{\mu}_t}{\sqrt{w_t^\top \hat{\Sigma}_t w_t}} - \lambda_c \sum_{m=1}^{M} w_{m,t}^2 \right\}, \tag{41}$$

with $\lambda_c = 0.25$. The penalty is small when weights are balanced and larger when the optimizer concentrates in one sleeve, so it acts as a soft diversification rule rather than a hard allocation cap.

Both portfolio methods are also volatility targeted. In this project, $\sigma^\star = 0.10$, so the target is 10% annualized volatility. The realized portfolio volatility estimate uses a 63-day rolling window and is annualized by 252. If $\sigma^\star$ is the target annualized volatility, then the leverage multiplier is

$$\lambda_t = \frac{\sigma^\star}{\hat{\sigma}_t^{\text{portfolio}}}, \tag{42}$$

and the targeted return uses lagged scaling:

$$r_t^{\text{targeted}} = \lambda_{t-1} r_t^{\text{portfolio}}. \tag{43}$$

The leverage multiplier is clipped between 0 and 2 so that the volatility target cannot create extreme exposure when recent realized volatility is very low.

Metric	Inverse Vol. IS	Markowitz Base IS	Markowitz Reg. IS
Sharpe ratio	1.21	1.17	1.20
Sortino ratio	1.54	1.37	1.45
CAGR (%)	13.00	12.67	12.83
Max drawdown (%)	-10.44	-9.37	-10.50
Calmar ratio	1.24	1.35	1.22
Volatility (%)	10.53	10.69	10.51
VaR_{95} (%)	0.98	1.00	1.00
ES_{95} (%)	1.57	1.69	1.62

Table 5: Selected portfolio methods before turnover caps and trading costs. All methods are volatility targeted; the regularized Markowitz column uses the concentration penalty in the objective.

Before execution constraints, the selected methods are close rather than decisively separated. Inverse-volatility has the highest Sharpe ratio (1.21) and CAGR (13.00%), because it gives meaningful weight to the defensive regime sleeve without abandoning the growth-oriented volatility-timing sleeve. The Markowitz baseline has the smallest drawdown (-9.37%) and strongest Calmar ratio (1.35), but its tail loss is higher, with ES_{95} of 1.69%. Regularization improves the baseline Sharpe from 1.17 to 1.20 and reduces ES_{95} from 1.69% to 1.62%, showing that a mild concentration penalty can stabilize the optimizer without replacing it with a fixed rule.

6.3 Execution Constraints

The realistic portfolio layer applies a turnover cap,

$$\tau_t \leq \tau^{\max}, \tag{44}$$

and charges the same transaction cost and slippage model used in the strategy backtests. Here $\tau^{\max} = 0.35$, so if the desired aggregate absolute weight change exceeds 0.35 on a rebalance

date, the whole change vector is scaled down to fit the cap. The realistic portfolio returns also use the same 5 basis points transaction cost plus 2 basis points slippage, for 7 basis points total per unit of turnover. This step is crucial because an optimizer that trades too aggressively can look good before costs but lose its advantage after implementation frictions.

	Inverse Vol.		Markowitz Base		Markowitz Reg.	
Metric	IS	OOS	IS	OOS	IS	OOS
Sharpe ratio	1.25	0.79	0.99	0.85	1.11	0.84
Sortino ratio	1.51	0.87	1.16	0.99	1.35	0.99
CAGR (%)	13.49	8.23	10.45	8.86	11.80	8.68
Max drawdown (%)	-10.47	-17.51	-11.46	-17.13	-11.30	-13.57
Calmar ratio	1.29	0.47	0.91	0.52	1.04	0.64
Volatility (%)	10.55	10.78	10.67	10.73	10.50	10.65
VaR_{95} (%)	0.94	0.96	1.05	1.00	1.02	0.99
ES_{95} (%)	1.60	1.71	1.70	1.70	1.62	1.68

Table 6: After-cost portfolio comparison with in-sample (IS) and walk-forward out-of-sample (OOS) results side by side. Returns include turnover caps, transaction costs, slippage, and volatility targeting; IS columns are aligned to the OOS evaluation calendar.

Table 6 compares the selected portfolios after execution realism. In-sample, inverse-volatility is strongest, with Sharpe 1.25, CAGR 13.49%, and max drawdown -10.47%. Its advantage comes from being a slow-moving risk rule: it does not estimate expected returns, so it is less exposed to noisy mean forecasts and expensive reallocations. Out of sample, all three methods weaken, which is expected once signals and weights are regenerated through time. Inverse-volatility keeps a positive OOS profile but its drawdown widens to -17.51%, showing that a simple risk rule can still remain exposed during equity stress. Markowitz baseline is competitive OOS, with Sharpe 0.85 and CAGR 8.86%, but it also suffers a large drawdown of -17.13%. The regularized Markowitz portfolio has similar OOS growth (8.68% CAGR) but the smallest OOS drawdown (-13.57%), which is why the concentration penalty is useful: it does not guarantee the highest return, but it makes the optimizer less fragile.

Fig. 7 summarizes the after-cost in-sample tradeoff across the three selected portfolios, while Fig. 8 shows the walk-forward OOS equity, drawdown, and rolling-Sharpe paths after costs. From Fig. 7 inverse-volatility and Markowitz baseline grow faster at times but experience deeper stress-period drawdowns, while regularized Markowitz has a smoother drawdown profile because the concentration penalty discourages extreme reallocations.

The full variant comparison in Appendix Tables A1 and A2, together with Appendix Figure A1, adds nuance. Ledoit–Wolf shrinkage alone barely changes the baseline, which suggests that covariance noise is not the main weakness in this two-sleeve portfolio. The 70% cap and shrinkage-plus-cap variants perform better than the unconstrained baseline because the cap prevents the

optimizer from leaning too heavily into one sleeve, but the cap is also an arbitrary hard boundary. The regularized Markowitz version is chosen as the main regularized comparison because it addresses the same concentration problem more smoothly: it discourages extreme weights inside the objective rather than forbidding them with a fixed ceiling. It is not selected because it is the best ex-post variant in every metric; it is selected because it is the cleanest regularized Markowitz benchmark for judging whether a softer optimizer can close the gap with inverse-volatility weighting.

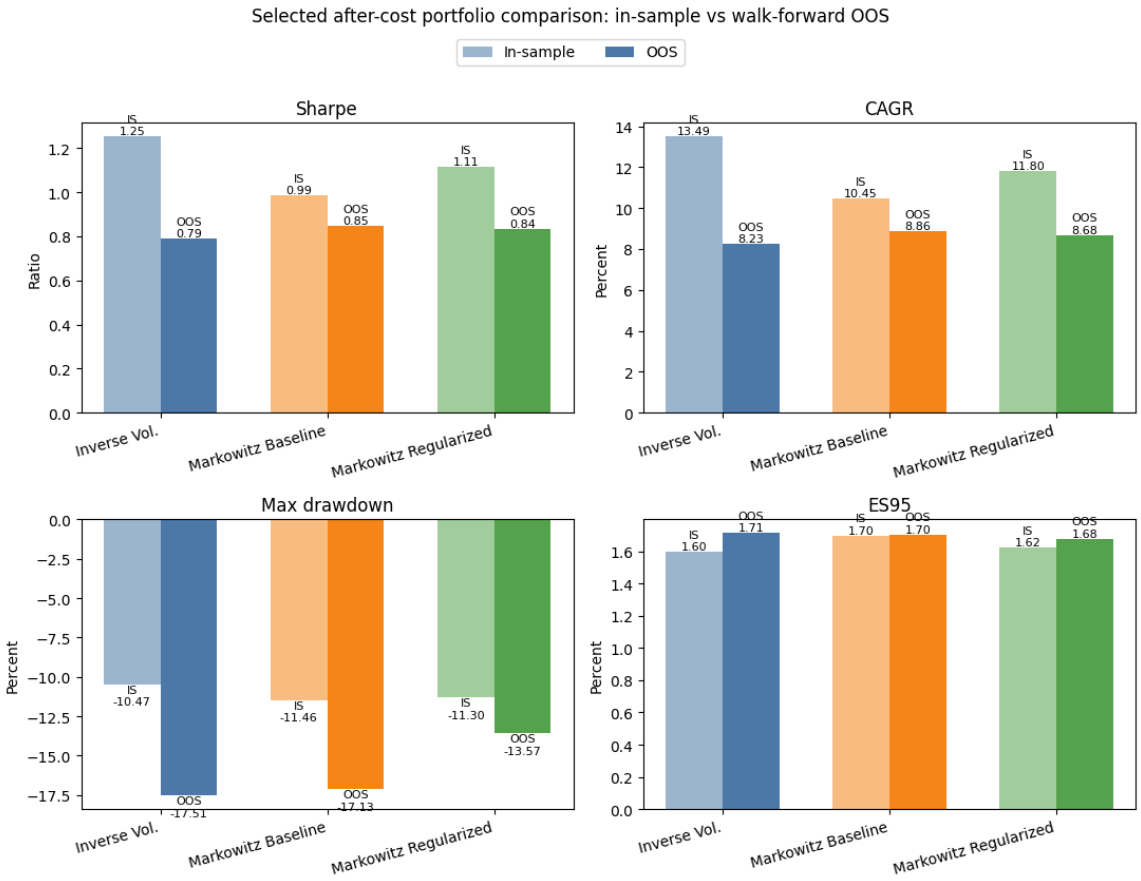


Figure 7: Selected after-cost in-sample portfolio comparison. Inverse-volatility has the strongest overall tradeoff, while concentration regularization improves Markowitz baseline performance but does not fully close the gap.

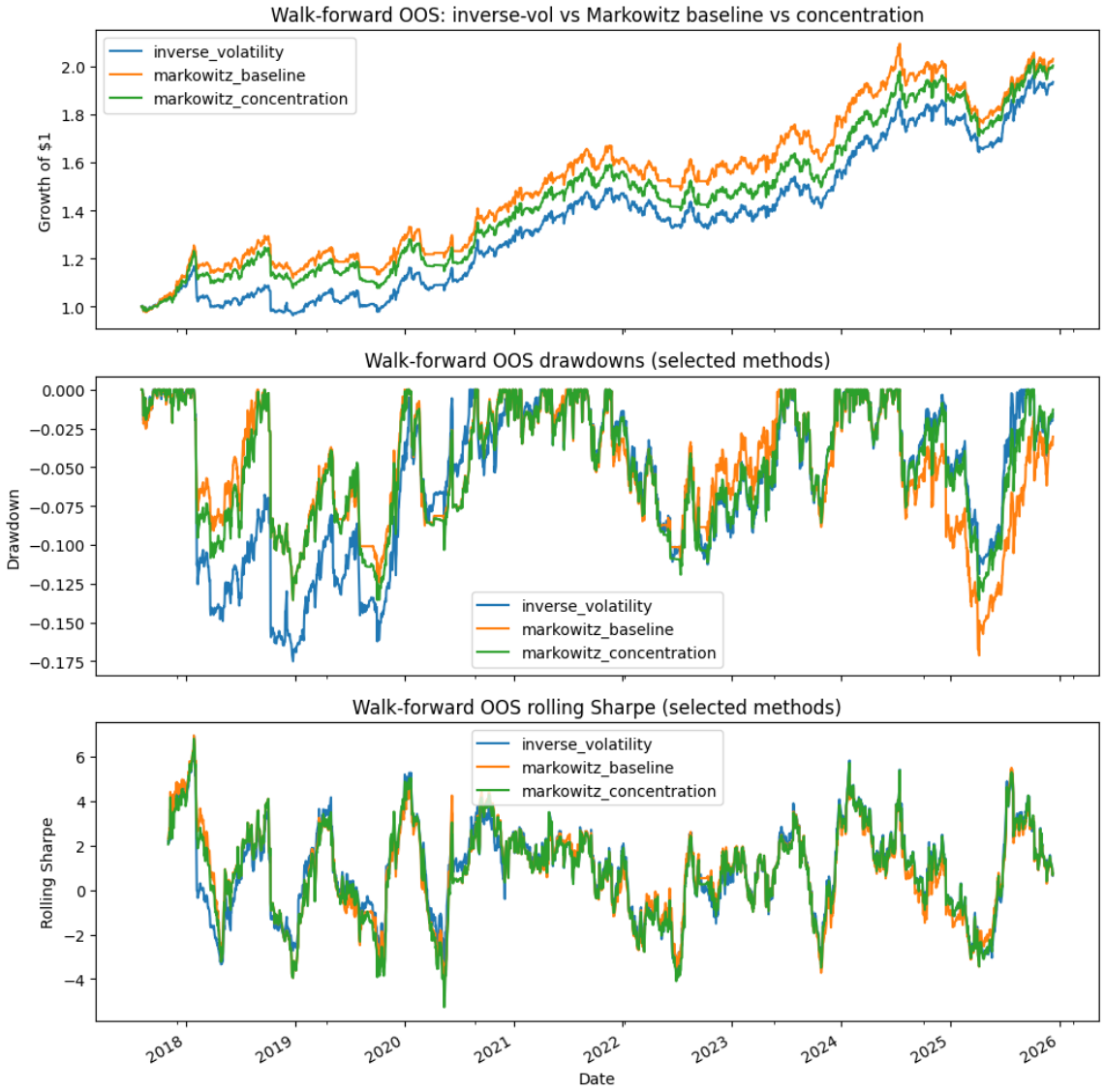


Figure 8: Walk-forward out-of-sample portfolio equity, drawdown, and rolling-Sharpe curves after turnover caps, transaction costs, slippage, and volatility targeting for the three selected portfolio methods.

7 Limitations

Several limitations should temper the interpretation of these results. The analysis uses adjusted daily ETF prices, which are appropriate for a transparent research prototype but do not capture intraday liquidity, order-book depth, borrow constraints, tax effects, or fund-level implementation frictions. The execution model applies fixed transaction-cost and slippage assumptions, so it approximates trading drag without modeling market impact, spread variation, capacity, or the possibility that costs rise during stressed regimes.

The modeling choices are also deliberately parsimonious. Strategy thresholds, lookback windows, volatility targets, and portfolio constraints are fixed rather than optimized across many market environments. The HMM regimes are useful statistical summaries, not stable economic states, and the random forest and Markowitz optimizer remain vulnerable to estimation error if retrained or tuned too aggressively. Finally, all reported metrics are historical estimates from one sample path. Sharpe ratio, CAGR, drawdown, VaR, and Expected Shortfall describe realized evidence, not guarantees about future performance.

8 Summary & Conclusion

This project demonstrates a complete quantitative research workflow: begin with clean data and testable hypotheses, translate only the credible signals into trading rules, and judge the final portfolio after costs, turnover limits, and risk controls. The central result is not that every model becomes a profitable strategy, but that each layer of the process asks a different question. A statistical signal must still survive trading rules, and a successful strategy must still improve a portfolio once implementation realism is included.

The evidence is clearest for risk management. Volatility forecasts from GARCH and random forest models contain useful information, and the volatility-timing strategy improves the return path relative to naive buy-and-hold. The HMM regime model also adds defensive value by reducing volatility, drawdown, and tail losses, although it gives up some upside in calmer markets. By contrast, the EFA–EEM pairs trade is rejected by both the cointegration test and the backtest, which is an important negative finding rather than a failure of the research process.

At the portfolio level, the strongest lesson is that robust allocation rules are often more reliable than fragile optimization. Inverse-volatility weighting delivers the best selected after-cost tradeoff, while Markowitz variants need caps or concentration penalties to control noisy reallocations. Overall, the project supports a disciplined view of systematic investing: useful structure exists in markets, but it should be trusted only after it has passed through statistical testing, trading simulation, portfolio construction, and realistic execution analysis.

Appendix A: Full Portfolio Variant Diagnostics

Metric	Inv. Vol.	Baseline	Shrinkage	70% Cap	Shrink.+Cap	Regularized
Sharpe ratio	1.21	1.17	1.17	1.22	1.23	1.20
Sortino ratio	1.54	1.37	1.37	1.49	1.50	1.45
CAGR (%)	13.00	12.67	12.66	13.12	13.25	12.83
Max drawdown (%)	-10.44	-9.37	-9.36	-10.18	-10.16	-10.50
Calmar ratio	1.24	1.35	1.35	1.29	1.30	1.22
Volatility (%)	10.53	10.69	10.69	10.52	10.53	10.51
VaR_{95} (%)	0.98	1.00	1.00	0.98	0.99	1.00
ES_{95} (%)	1.57	1.69	1.68	1.62	1.62	1.62

Table A1: Full in-sample portfolio variant comparison before turnover caps, transaction costs, and slippage. All variants are volatility targeted; the main text reports the inverse-volatility, Markowitz baseline, and concentration-regularized Markowitz columns.

	Inv. Vol.		Baseline		Shrinkage		70% Cap		Shrink.+Cap		Regularized	
Metric	IS	OOS	IS	OOS	IS	OOS	IS	OOS	IS	OOS	IS	OOS
Sharpe ratio	1.25	0.79	0.99	0.85	0.98	0.85	1.16	0.84	1.16	0.84	1.11	0.84
Sortino ratio	1.51	0.87	1.16	0.99	1.15	1.01	1.41	0.97	1.40	0.98	1.35	0.99
CAGR (%)	13.49	8.23	10.45	8.86	10.36	8.94	12.37	8.71	12.33	8.76	11.80	8.68
Max drawdown (%)	-10.47	-17.51	-11.46	-17.13	-11.45	-16.27	-11.08	-14.42	-11.07	-14.15	-11.30	-13.57
Calmar ratio	1.29	0.47	0.91	0.52	0.91	0.55	1.12	0.60	1.11	0.62	1.04	0.64
Volatility (%)	10.55	10.78	10.67	10.73	10.68	10.72	10.51	10.69	10.54	10.69	10.50	10.65
VaR_{95} (%)	0.94	0.96	1.05	1.00	1.06	1.00	0.99	0.98	0.99	0.98	1.02	0.99
ES_{95} (%)	1.60	1.71	1.70	1.70	1.69	1.70	1.63	1.70	1.63	1.70	1.62	1.68

Table A2: Full after-cost portfolio variant comparison with in-sample (IS) and walk-forward out-of-sample (OOS) results side by side. Returns include turnover caps, transaction costs, slippage, and volatility targeting; IS columns are aligned to the OOS evaluation calendar.

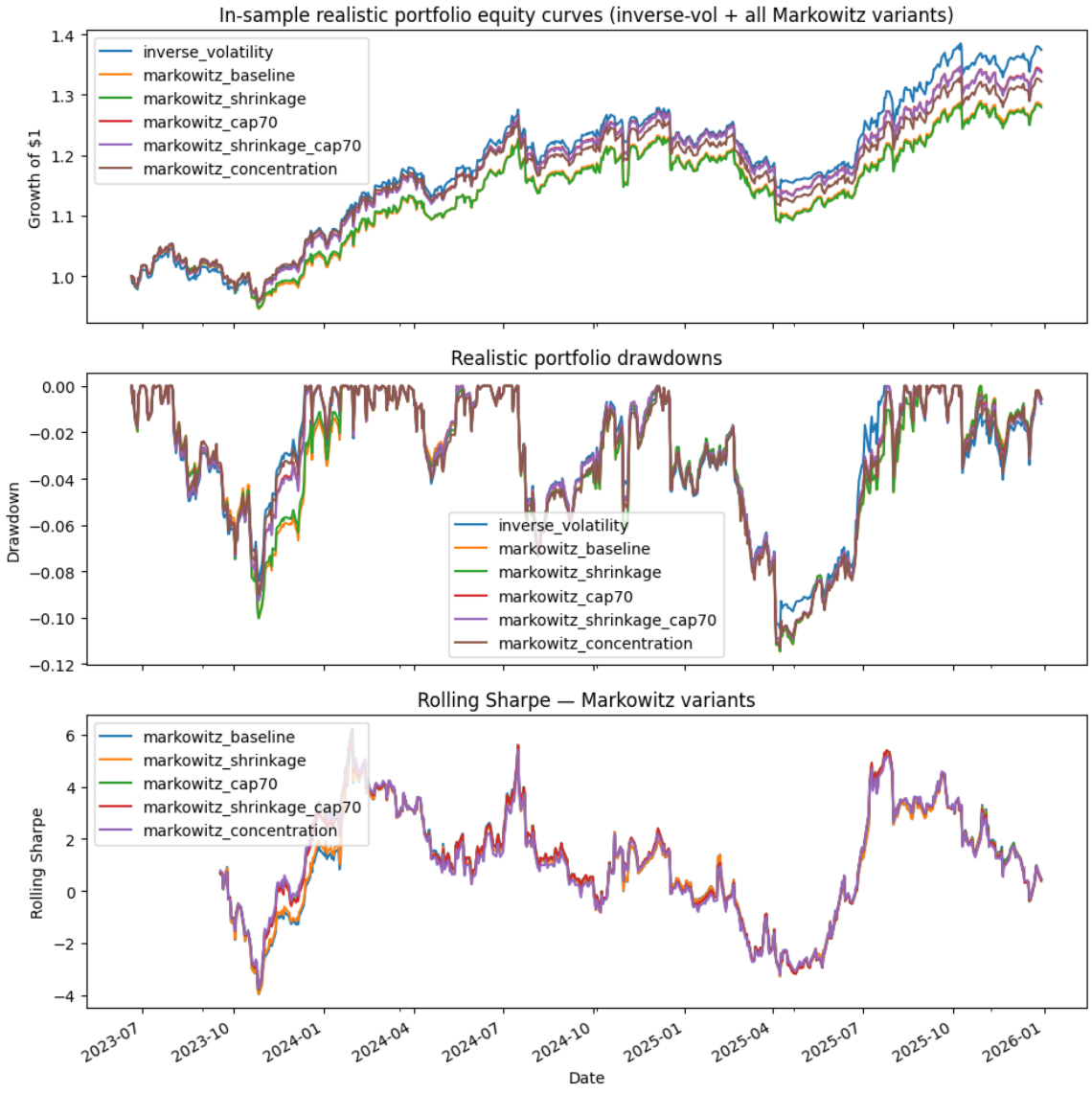


Figure A1: Full realistic portfolio equity, drawdown, and rolling-Sharpe diagnostics for inverse-volatility and all Markowitz variants.